N-digits Ternary Carry Lookahead Adder Design

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***Abstract*—Carry lookahead adders (CLAs) are extensively used in digital circuits due to their logarithmic computational time (O(log n)) compared to linear computational time(O(n)) in the ripple carry adders. In this paper, two design approaches for N-digits ternary logic CLA based on K-map and threshold logic methods are proposed in addtion to their realization using CNTFETs only and memristor with CNTFETs. Finally, 4-bit ternary CLA is presented. A comparison and tradeoffs among the proposed designs are presented in terms of the delay and the area. The comparison shows that the transistor-only-based implementation is the best choice in the K-map design approach. However, the memristor and transistor-based implementation based on memristor and transistors integration is the best in the threshold logic (TL) design. Therefore, the proposed designs are very promising to build high performance full ternary ALU unit.**

keywords – Ternary Numbers, Threshold Logic, Memristor, CLA, CNTFET.

I. Introduction

Ternary numbers which has three logic values, attracted the researcher attention for its proximity base to the natural logarithm e [1]. Its odd valued logic facilities representing rational and complex number rather than binary number [2]. In addition, its base provides more information to be represented in the same number of digits by 1.5^N compared with binary numbers. These superior properties are stimulated to improve computational performance and shine in the fields of artificial intelligence logic [3], cryptography systems [4], communication [5], graphical processing [6] and fuzzy logic systems [7]. In in 1950s, it was the first trial to build a ternary computer emulator by Russians [8]. Later, researchers commuted from emulators design to implement the first arithmetic operations from the circuit level based on transistors in 1980's [9].

The nano-scale dimensions of the memristors (RRAMs) enabled many applications specially analog applications due to their analog behavior [11]. One of the uprising concepts based on memristors is the memristive threshold logic (MTL), inspired by the synaptic of the neuron in the brain [12]. It opened the space for speeding up and minimizing the area of many circuits [12], [14]. MTL is emerged in many applications such as logic gates, programmable logic array, fast switching memory, reconfigurable computing, pattern recognition, image processing, and biological brain neuron network [13], [14].

Recently, the researchers tried to implement ternary numbers functions based on the integration between transistors and memristor [15]–[18]. In [15], the ternary numbers logic gates have been implemented using memristor integrated with CNTFETs to build one of the ternary arithmetic operation. Two methods were introduced: the first one was based on the conventional K-map design. The second one was based on the memristor threshold-adaption method which was introduced for the first time to improve the overall circuit performance [15]. However, these designs have not been used to implement fast N-bits arithmetic operations, yet and accordingly, building high performance full ternary ALU unit.

One of the most common fast adders in binary systems is the carry-lookahead adder (CLA). It was introduced for the first time in binary digital logic by Rosenberger [19]. It is called a fast or parallel adder where the propagated carries are computed in parallel without waiting for the sum of digits sequence [19]. That's why, it has been emerged in the modern computers. On the side of the ternary number, the previous work has discussed only how to realize the ternary CLA [5], [20] but, there are no logic circuits implementations, yet.

In this paper, novel designs for carry lookahead adder are proposed and used to build N-digits CLA adder. K-map and threshold logic design methods, presented in [15], are used. A general comparison for the proposed design are presented using transistor technology and the technology of the integration between memristor with transistor as well. The pros and cons of each tecnologies are discussed. Finally, the implementation of 4-bit ternary CLA is introduced.

This paper is organized as follows: Section II introduces different designs for ternary carry lookahead adder components. Section III shows how to use the proposed design to build N-bit CLA adder. Section IV presents the simulation results and general comparison for the proposed designs using different technologies. The conclusion is drawn in Section V.

II. 1-bit ternary carry-lookahead adder

In CLA, the carry is defined by two signals; either generated (g) as an output or propagated (p) as an input to the next digit in the input sequence. The carry function is defined as follows:

$$c_{i+1} = g_i + p_i.c_i, \tag{1}$$

where i is the number of current digit. In ternary systems, The same concept is utilized where the carry is always either logic 0 or 1. Since g and p are different states of the carry, they are either logic 0 or 1. Table I shows the g and p truth table of two inputs A and B where x is the do not care output. The g

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and p functions are generalized in ternary logic to $g = carry$ and $p = 1$ whole the sum is logic 2.

TABLE I: p and g truth table

	p			g		
A/B	0	1	2	0	1	2
0	0	0	1	0	0	0
1	0	1	x	0	0	1
2	1	x	x	0	1	1

A. p and g Design using K-map Approach

According to the truth table shown in Table I, p and g functions can be deduced using the K-map. By considering the do not care p outputs with logic 1, $p = 1.(A_2B_0 + A_2B_1 + A_2B_2) + 1.(B_2A_0 + B_2A_1 + B_2A_2) + 1.A_1B_1$ which can be reduced to

$$p = 1.(A_2 + B_2 + A_1B_1) \quad (2a)$$

$$g = 1.(A_1B_2 + A_2B_1 + A_2B_2) \quad (2b)$$

where $A_0 + A_1 + A_2 = 2$ and $B_0 + B_1 + B_2 = 2$, $A_2.2 = A_2$ and $B_2.2 = B_2$. Figure 1 shows the logic circuit designs of g and p functions in (2) based on K-map.

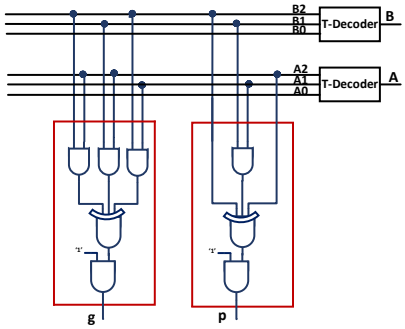


Fig. 1: MTL design general architecture.

B. p and g Design using Threshold Logic (TL)

The design of g and p functions using (TL) is mainly based on finding the relationships between the logic gates under specific logic conditions.

The logic circuit of the g function can be defined as follows:

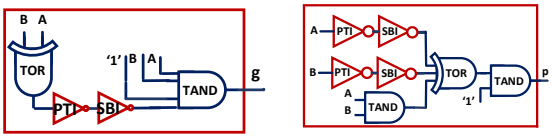
- When $TOR(A, B)$ equals 2, g output is almost the same as the output of $TAND(A, B)$ as shown in Table II, except the case of (A, B) = $(2, 2)$.
- For appropriate the case of (A, B) = $(2, 2)$ with the output of g function, the output of $TAND(A, B)$ is down converted to logic 1 by ANDing this output with logic 1. Therefore, the function $TAND(A, B, 1)$ describe the g function under the condition $TOR(A, B) = 2$

Figure 2(a) shows the logic circuit of g function. The binary inverter (SBI) and ternary positive inverter (PTI) which described in [15], are used to check on either the logic output is 2 or not as follows: when the condition $TOR(A, B) = 2$ is achieved, the output of $SBI(PTI(TOR(A, B))) = 2$ and otherwise $SBI(PTI(TOR(A, B))) = 0$. Finally, the condition $TOR(A, B) = 2$ and the function $TAND(A, B, 1)$ are ANDed together as shown in Fig.2(a).

The p function which defined by $(p = A_2 + B_2 + A_1B_1)$, can be described using TL as follows:

TABLE II: 1-digit CLA truth table

A	B	p	g	Sum	$Carry$	TOR	$TAND$
0	0	0	0	0	0	0	0
0	1	0	0	1	0	1	0
0	2	1	0	2	0	2	0
1	0	0	0	1	0	1	0
1	1	1	0	2	0	1	1
1	2	x	1	0	1	2	1
2	0	1	0	2	0	2	0
2	1	x	1	0	1	2	1
2	2	x	1	1	1	2	2



(a) (b)

Fig. 2: TL based Logic circuit for (a) p, and (b) g function.

- The terms A_2 and B_2 which depicted to $A = 2$ and $B = 2$, can be checked by $SBI(PTI(A))$, $SBI(PTI(B))$, respectively. When $A = 2$ or $B = 2$, the $SBI(PTI(A)) = 2$ or $SBI(PTI(B)) = 2$ and otherwise, $SBI(PTI(A)) = 0$ or $SBI(PTI(B)) = 0$, respectively.
- The term A_1B_1 can be obtained by the function $TAND(A, B)$ under the condition $TOR(A, B)! = 2$ as shown in Table II.

Figure 2(b) shows the logic circuit of p function where the functions $SBI(PTI(A))$, $SBI(PTI(B))$ and $TAND(A, B)$ are ORed together. Finally, to guarantee the maximum output of p function is logic 1, the output of the ORed terms are ANDed with logic 1.

III. N-BIT TERNARY CARRY-LOOKAHEAD ADDER DESIGN

The procedure to construct N-digits CLA can be summarized in the following steps:

- Firstly, the g_0, p_0, g_1, p_1,..., g_n, p_n are obtained directly from the adder inputs A_0, B_0, A_1, B_1,..., A_n, B_n.
- Then, all carries are obtained in parallel (at the same time) as shown in Fig.3(a).
- Finally, the sum phase is performed as shown in Fig.3(b) where the parallel carries are added with the inputs sequence using ternary full adder (T-FA).

On the other hand, the ripple carry adder needs to N-stages for the N-bit adder. However, increasing numbers of digits in the CLA architecture shown in Fig.3(b), may cause a problem called fan-in problem. The fan-in in digital circuits is defined

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TABLE III: g and p comparison between different design technologies

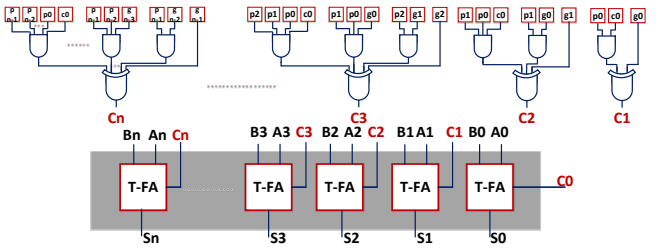
	Approach	Path Gate Delay	Area (g and p)	
			#mem	#trans
K-map	transistors only	D_g=D_p=2 D_{SBI}+ D_{OR_2}+D_{AND_2}+D_{OR_3}+D_{TAND_2}	–	92
	memristors & transistors	D_g=D_p=8D_{SBI}+D_{TOR_2} + 2D_{TAND_2}+D_{TOR_3}	22	52
TL	transistors only	$D_p = \max\{2D_{SBI} + D_{TOR_3} + D_{TAND_2}, D_{TOR3} + 2D_{TAND2}\}$ $D_g = 2D_{SBI} + D_{TOR_2} + D_{TAND_4}$	–	104
	memristors & transistors	$D_p = \max\{4D_{SBI} + D_{TOR_3} + D_{TAND_2}, 2D_{SBI} + 2D_{TAND_2} + D_{TOR_3}\}$ $D_g = 2D_{SBI} + D_{TOR_2} + D_{TAND_4}$	13	36

TABLE IV: Logic gate area in general form using different design technologies

		trans.only	mem.+trans	
		#trans.	#mem.	#trans.
Binary Gates	SBI	2	–	2
	OR_2/AND_2	6	2	4
	OR_3/AND_3	8	3	4
Ternary Gates	$TOR_2/TAND_2$	16	2	6
	$TOR_3/TAND_3$	20	3	6
	$TOR_4/TAND_4$	24	4	6

$$c_1 = g_0 + p_0c_0$$
$$c_2 = g_1 + p_1g_0 + p_1p_0c_0$$
$$c_3 = g_2 + p_2g_1 + p_2p_1g_0 + p_2p_1p_0c_0$$
$$c_4 = g_3 + p_3g_2 + p_3p_2g_1 + p_3p_2p_1g_0 + p_3p_2p_1p_0c_0$$
$$\vdots$$
$$c_n = g_{n-1} + p_{n-1}g_{n-2} + p_{n-1}p_{n-2}g_{n-3} + \cdots + p_{n-1}p_{n-2}\ldots g_0 + p_{n-1}p_{n-2}\ldots c_0$$

(a)



(b)

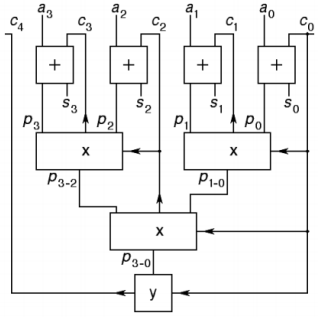
Fig. 3: The carry Lookahead adder general (a) definition, and (b) architecture.

as the maximum number of inputs that a logic gate can accept. If number of input exceeds, the output may be undefined or incorrect. Therefore, the general architecture, shown in Fig.3(b), has been modified to the hierarchical ternary CLA shown in Fig.4(b) to avoid this problem. Figure 4(a) shows the modified terms in carries form where c_4 is re-defined by new $G_{[0:3]}$ and $P_{[0:3]}$ functions and similarly to the next carries. In the optimized CLA design, extra stages of carry lookahead generator block (x) is added on the design to eliminate the propagated terms in carries form.

The carry generator block is divided into number of (x) blocks with n-bit of carries outputs as shown in Fig.4(b). Each block has its own initial carry input which is generated by the previous carry lookahead generator block. Each block adds one extra stage to eliminate the fan-in gate problem. Finally, after all carries are obtained, they propagate to the full adders stage (y) to generate the final sum outputs as shown in Fig.4(b).

$$c_1 = g_0 + p_0c_0$$
$$c_2 = g_1 + p_1g_0 + p_1p_0c_0$$
$$c_3 = g_2 + p_2g_1 + p_2p_1g_0 + p_2p_1p_0c_0$$
$$c_4 = g_3 + p_3g_2 + p_3p_2g_1 + p_3p_2p_1g_0 + p_3p_2p_1p_0c_0$$
$$c_4 = G_{[0:3]} + P_{[0:3]}c_0$$
$$c_5 = g_5 + p_5\,G_{[0:3]} + p_5\,P_{[0:3]}c_0$$
$$\vdots$$

(a)



(b)

Fig. 4: The carry Lookahead adder optimized (a) definition, and (b) architecture.

IV. The Comparison and Simulation Results

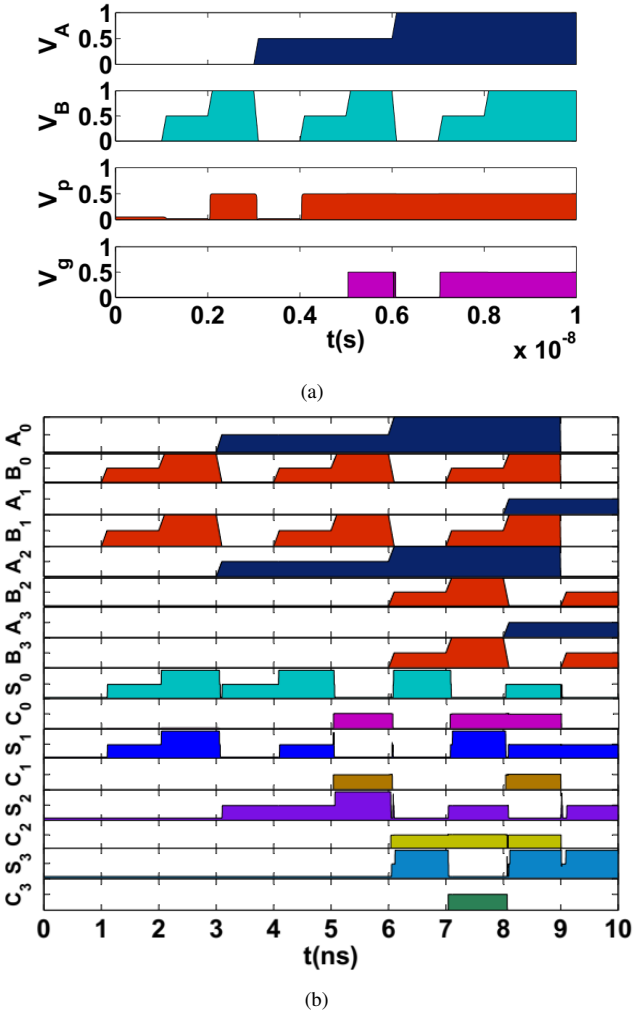
The proposed p and g functions designs have been implemented using different two technologies; transistors only and the integration between memristors and transistors. Table III shows a comparison between the proposed K-map and TL design approaches of p and g in terms of the estimated area and delay. The following is the summary of the findings.

- The area in K-map design using transistors only is less than TL-based design where TL-based design is built based on ternary gates only. While K-map design is build based on binary logic gates except the last stage which uses ternary logic gates to convert the output to ternary number. As shown in Table IV, ternary logic gates have larger area than binary gates in case of the implementation using transistors only.
- The implementation using memristors with transistors has less area compared with using the transistors only.

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In general, the delay differs from technology to another, so general expressions for delay paths are reported. Table III presents general delay forms for the possible paths in the proposed designs where the D_{OR_2} and D_{OR_3} are the delay of the two and three bit OR gate, respectively.

Figure 5(a) shows SPICE simulation results of g and p for TL-based design. It is simulated based on the integrated technology between memristor and transistors where VTEAM memristor and CNTFET transistor models are used [21], [22]. The following memristor paramters are used in simulations; $R_{on} = 50\Omega$,$R_{off} = 1k\Omega$, $v_{th_1} = -0.02v, v_{th_2} = 0.02v$, no window is employed and with a linear I-V relationship. The transistor model of Stanford Virtual-Source Carbon Nano-tube (VS CNTFET) is used [22]. The simulation results shows that delay of $D_p = 86.64ps$ and $D_g = 45.42ps$. Finally, they are used with the half adder represented in [15] to implement 4-bit ternary CLA as depicted in the simulation result shown in Fig.5(b).



(a)

(b)

Fig. 5: The simulation result of (a) the p and g functions, and (b) 4-bit CLA adder using VTEAM memristor, and CNTFET transistor models.

V. Conclusion

In this paper, novel designs for ternary carry lookahead adder components have been introduced using different design approaches. The proposed designs have been used to build the N-bit ternary CLA. A comparison between the proposed designs using different technologies have been presented. It shows that in the case of implementing using transistors only, the area is K-map design is better than TL. In case of implementing using the integration between memristor and transistor technology, the TL design reduced the number of transistors over the K-map design. Consequently, the TL design is not only has better area but also static power than K-map design. The proposed designs for CLA are very promising to be emerged to build high performance full ternary ALU unit based on different technologies.

References

[1] B. Hayes, Third Base, American Scientist, vol. 89, no. 6, p. 490, 2001.
[2] D. Dao, et al., Complex Number Arithmetic with Odd- Valued Logic, IEEE Transactions on Computers, vol. C29, no. 7, pp. 604611, Jul. 1980.
[3] S. N. Yanushkevich, Editorial Artificial Intelligence in Logic Design, Artificial Intelligence in Logic Design, pp. 78, 2004.
[4] J. Adikari, et al., Hybrid Binary-Ternary Number System for Elliptic Curve Cryptosystems, IEEE Transactions on Computers, vol. 60, no. 2, pp. 254265, Feb. 2011.
[5] B. Parhami, et al., Arithmetic with binary-encoded balanced ternary numbers, 2013 Asilomar Conference on Signals, Systems and Computers, Nov. 2013.
[6] E. A. Weiss, In the Art of Programming, Knuth Is First; There Is No Second, A Computer Science Reader, pp. 282291, 1988.
[7] V. Novk, Which logic is the real fuzzy logic?, Fuzzy Sets and Systems, vol. 157, no. 5, pp. 635641, Mar. 2006.
[8] D. Knuth, "The Art of Computer Programming," Second edition. Addison Wesley Publishing Company, 1981.
[9] C.-Y. Wu, et al., "Design and application of pipelined dynamic CMOS ternary logic and simple ternary differential logic," IEEE J. Solid-State Circuits, vol. 28, no. 8, pp. 895-906, 1993.
[10] A. G. Radwan, et al., "On the Mathematical Modeling of Memristor, Memcapacitor, and Meminductor," Studies in Systems, Decision and Control, 2015.
[11] A. P. James, "The Memristor Circuits and Applications," Advances in Memristor Circuits and Bioinspired Systems, 2015.
[12] A. K. Maan, et al., "A Survey of Memristive Threshold Logic Circuits," IEEE Trans. Neural Netw. Learn. Syst., vol. 28(8), pp. 1734-1746, 2017.
[13] R. Kozma, et al., "Advances in Neuromorphic Memristor Science and Applications," Springer, 2012.
[14] P. Mazumder, et al., "Memristors: Devices, models, and applications," Proc. IEEE , vol. 100, no. 6, pp. 1911-1919, Jun. 2012.
[15] N. S. Soliman, et al., "Memristor-CNTFET based ternary logic gates," Microelectronics Journal, vol. 72, pp. 74-85, Feb. 2018.
[16] M. Ajmal, et al., Reconfigurable Memristor and CNFET based Four Quadrant Multiplier for Low Power Applications, International Journal of Computer Applications, vol. 173, no. 6, pp. 1420, Sep. 2017.
[17] N. S. Soliman, et al., "Memristor-CNTFET based Ternary Comparator unit," IEEE International Conference on Microelectronics (ICM), Dec. 2018.
[18] N. Soliman, et al., "Ternary Functions Design Using Memristive Threshold Logic," IEEE Access, vol. 7, pp. 4837148381, 2019.
[19] V. Kantabutra, A recursive carry-lookahead/carry-select hybrid adder, IEEE Trans. on Computers, vol. 42, no. 12, pp. 14951499, 1993.
[20] M. H. A. Khan, et al., Quantum ternary parallel adder/subtractor with partially-look-ahead carry, Journal of Systems Architecture, vol. 53, no. 7, pp. 453464, Jul. 2007.
[21] S. Kvatinsky, et al. "VTEAM: A General Model for Voltage-Controlled Memristors," IEEE Trans. Circuits Syst. II, Exp. Briefs, vol. 62, no. 8, pp. 786-790, Aug. 2015.
[22] https://nano.stanford.edu/stanford-cnfet-model.

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